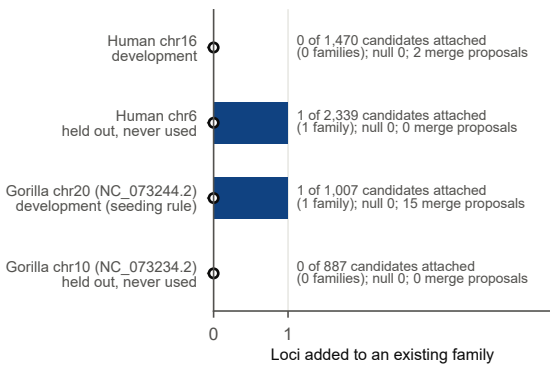
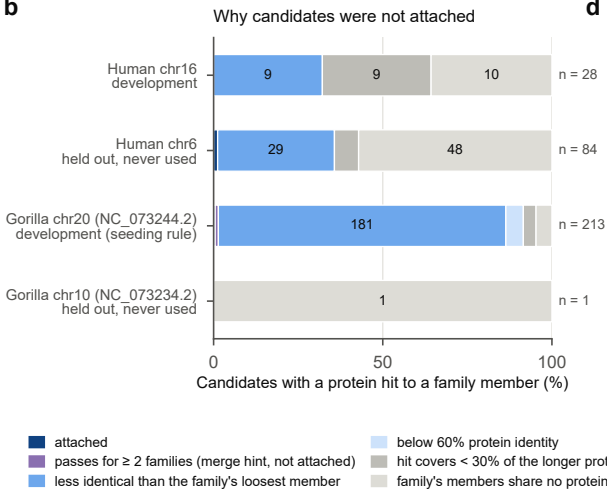


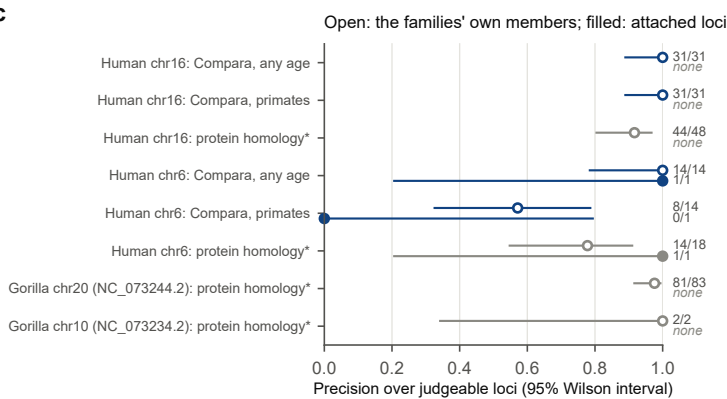
a What the step adds (circle: shuffled null)



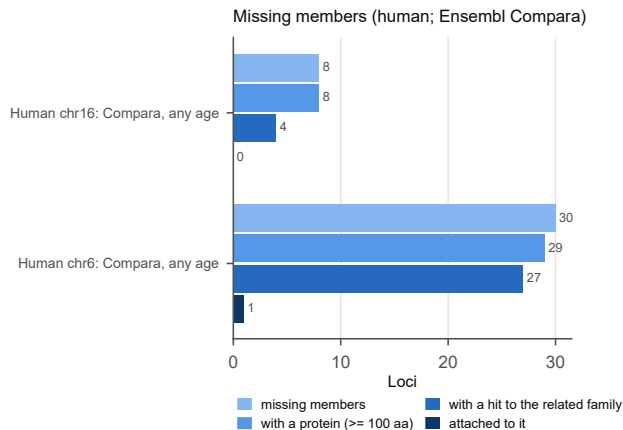
b



c



d



Protein is not part of the default family definition: this optional, manually run step adds a locus the RNA rule left out to one existing family and never merges two. Missing member: an unattached locus whose gene no family holds and which Compara relates to a member's gene. * Protein-homology families are built from protein homology, so agreement with them is partly by construction (secondary only). No missing member at the primates level (Human chr16, Compara, primates; Human chr6, Compara, primates). Not shown, Ensembl Compara, paralogues of any age and Ensembl Compara, duplications within primates: Compara export in hand is human only. Not shown, Liftoff copies (self-lift, $\geq 95\%$ identity): Liftoff self-lift not finished for this species.